
MINOR ASSIGNMENT-08

IPC & Process/Thread Synchronization

Practical Programming with C (CSE 3544)

Publish on: 21-12-2025

Course Outcome: CO₅

Program Outcome: PO₃

Submission on: 28-12-2025

Learning Level: L₅

Problem Statement

Experiment with inter-process communication mechanisms(IPCs) and semaphores

Answer the followings:

5. Write a program to create a shared memory segment of size 10 bytes. Make 4 attachments to the shared memory segment to the address space of the calling process and print the number of attachments using the structure field *number of current attachments* present in the structure **shmid_ds** defined in the header **<sys/shm.h>**. Check the number of attachment using the shell provided command **ipcs -m**.

Code here

Specify: input & output

6. Create a C code named **shmwriter.c** to create a shared memory segment of integer size and store 500 to the segment. Create another program named **shmreader.c** to access the stored value from the shared memory segment and display it. Let the **shmreader.c** update the value to 600. Now update the **shmwriter.c** code to get the updated value and display it. You are not allowed to use `semaphore`.

Code here

Specify: input & output

7. Create 2 processes using **fork()**. The child will send a number to parent using shared memory segment. The parent will display the received number and doubles it and sends back to the client. The client will display the received number.

Code here

Specify: input & output

8. Design a C code to create a message queue and add 4 messages to the queue. Create a receiver code to receive all the messages from the queue till the queue is empty.

Code here

Specify: input & output

9. Find the critical section and determine the race condition for the given pseudo code. Write program using **fork()** to realize the race condition(**Hint:** *case-1:* Execute P_1 first then P_2 , *case-2:* Execute P_2 first then P_1). Here **shrd** is a shared variables.

P-1 executes:

```
x = *shrd;
x = x + 1;
sleep(1);
*shrd = x;
```

P-2 executes

```
y = *shrd;
y = y - 1;
sleep(1);
*shrd = y;
```

Code here

10. Suppose **process 1** must execute statement **a** before **process 2** executes statement **b**. The semaphore **sync** enforces the ordering in the following pseudocode, provided that **sync** is initially 0.

Process 1 executes:

```
a;
signal(&sync);
```

Process 2 executes:

```
wait(&sync);
b;
```

Because **sync** is initially 0, **process 2** blocks on its **wait** until **process 1** calls **signal**. Now, You develop a C code using **POSIX: SEM semaphore** to get the desired result.

Code here

Code here

11. Implement the following pseudocode over the semaphores S and Q. State your answer on different initializations of S and Q.

(a) S=1, Q=1

(d) S=0, Q=0

(b) S=1, Q=0

(c) S=0, Q=1

(e) S=8, Q=0

Process 1 executes:

```
for( ; ; ) {  
    wait(&S);  
    a;  
    signal(&Q);  
}
```

Process 2 executes:

```
for( ; ; ) {  
    wait(&Q);  
    b;  
    signal(&S);  
}
```

Code here

12. Implement and test what happens in the following pseudocode if semaphores S and Q are both initialized to 1?

process 1 executes:

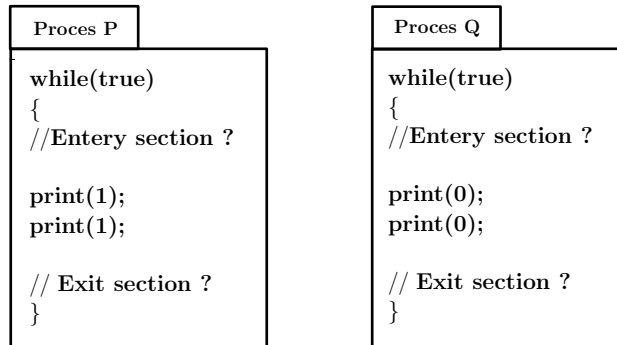
```
for( ; ; ) {  
    wait(&Q);  
    wait(&S);  
    a;  
    signal(&S);  
    signal(&Q);  
}
```

process 2 executes:

```
for( ; ; ) {  
    wait(&S);  
    wait(&Q);  
    b;  
    signal(&Q);  
    signal(&S);  
}
```

Code here

13. Two concurrent processes **P** and **Q** are accessing their critical sections by using boolean variables **S** and **T** as follows;



Design a code to complete the entry and exit section of process **P** and **Q** with appropriate semaphore operations using the two boolean semaphores **S** and **T**. Also suggest the initial values of **S** and **T**, such that the execution of the processes will print the sequence **00110011**

Code here

14. You have three processes/ threads as given in the below diagram. Create your C code to print the sequence given as;
XXYZZYXXYZZZYXXYZZYXXYZZZ. You are required to choose any one of the process/ thread synchronization mechanism(s).

Process/Thread -1

```
for i = 1 to ?  
:  
:  
display("X");  
display("X"); :  
:
```

Process/Thread -2

```
for i = 1 to ?  
:  
:  
display("Y");  
:  
:
```

Process/Thread -3

```
for i = 1 to ?  
:  
:  
display("Z");  
display("Z"); :  
:
```

Code here

15. **Process ordering using semaphore.** Create 6 number of processes (1 parent + 5 children) using **fork()** in **if-elseif-else** ladder. The parent process will be waiting for the termination of all it's children and each process will display a line of text on the standard error as;

```
P1: Distance
P2: KEEP
P3: Smart
P4: From
P5: Devices
```

Your program must display the message in the given order

Keep Distance From Smart Devices.

Code here

- **OPTIONAL** Write a program to give a semaphore-based solution to the producer-consumer problem using a bounded buffer scheme.

Code here

- **OPTIONAL The Sleeping-Barber Problem.** A barbershop consists of a waiting room with n chairs and a barber room with one barber chair. If there are no customers to be served, the barber goes to sleep. If a customer enters the barbershop and all chairs are occupied, then the customer leaves the shop. If the barber is busy but chairs are available, then the customer sits in one of the free chairs. If the barber is asleep, the customer wakes up the barber. **Write a program to coordinate the barber and the customers.**

Code here

Code here

- **OPTIONAL The Cigarette-Smokers Problem.** Consider a system with three smoker processes and one agent process. Each smoker continuously rolls a cigarette and then smokes it. But to roll and smoke a cigarette, the smoker needs three ingredients: tobacco, paper and matches. One of the smoker processes has paper, another has tobacco, and the third has matches. The agent has an infinite supply of all three materials. The agent places two of the ingredients on the table. The smoker who has the remaining ingredient then makes and smokes a cigarette, signaling the agent on completion. The agent then puts out another two of the three ingredients, and the cycle repeats. **Write a program to synchronize the agent and the smokers using any synchronization tool.**

Code here

Code here